
MICOS
RS 232 IN
STATUS
MOCO
RS 232 OUT

MoCo-*Series*

Motion made easy

User Manual for MoCo Controllers

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Key features of MoCo DC external

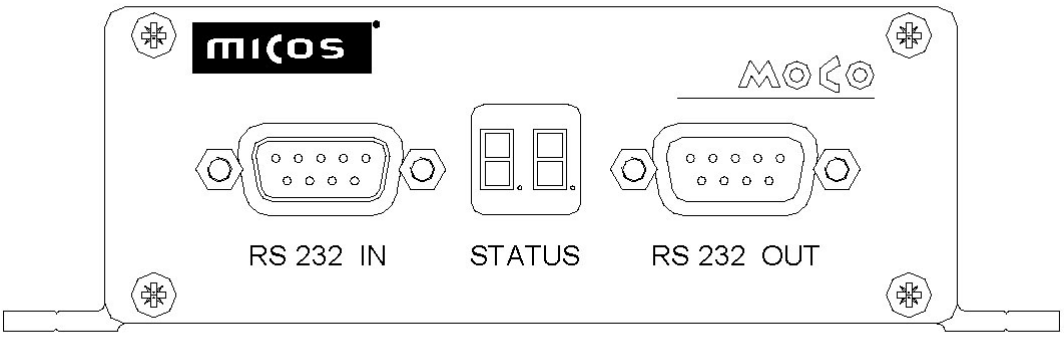
- LM629 Servo Controller for simultaneous control of position, velocity and acceleration, with PID filtering.
- LMD18201 Motor Driver (3 A continuous current, 6 A for 200 mSec)
- Non-volatile (permanent) storage of servo parameter setups and user programs.
- On-board control language with over 50 commands.
- Optically isolated, wide voltage range limit switch inputs (2).
- Single-ended or differential encoder inputs, up to 1 MHz, with index capability.
- Position Synchon Trigger Output (since V2.91)
- Limit Switch disable function (since V4.00 and HW Rev 610/611)

- Up to 16 MoCo controllers may share a single RS-232 channel, with individual addressing. Two paralleled connectors simplify multi-drop cabling.
- Large, two-digit LED for network address and error code display.

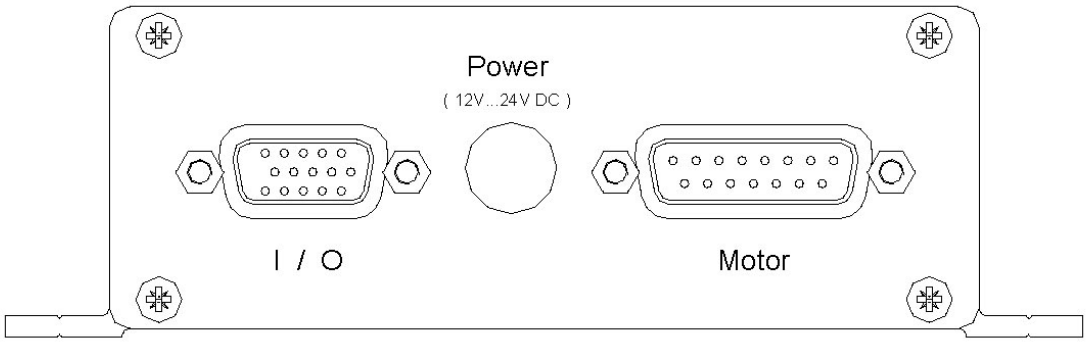
- Operation from a single power supply 12 .. 24 VDC, typically 24 VDC.
- Additional connector for general purpose input/output, fully supported by the integrated command structure.

- Two analog inputs, 0-5 V, 8 bit resolution
- One analog output, 0-5 V, 8 bit resolution, 2 mA.
- Four optically-isolated inputs.
- Four high-voltage, high-current driver outputs. Separate input for driver voltage supply.

Front View



Rear View



Connectors

X 1: Motor, Encoder & Limit

DB15f	Function		DB15f
1	A+	A-	9
2	B+	B-	10
3	C+	C-	11
4	Gnd	5 V	12
5		Brake-Out	13
6	M+	M-	14
7	E2 (limit forward)	E1 (limit reverse)	15
8	Limit-24VDC		

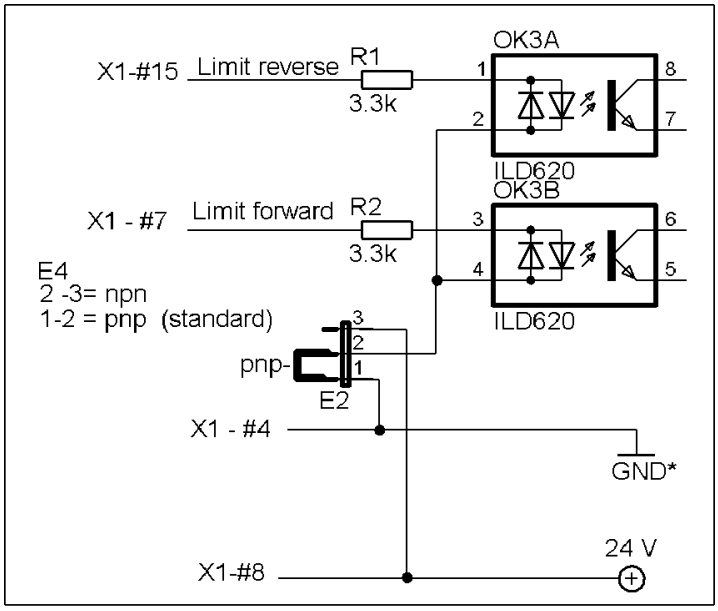
The limit-switch inputs are direction dependent. Limit E1 stops reverse motion, Limit E2 stops positive motion!

Do not connect or disconnect motor or power leads with power applied!

Interface :

The inputs could be configured as **pnp** or **npn** by internal jumper.

Default : **pnp** sourcing the input (connect to V+ (5..24VDC)
npn sinking the input (connect to Gnd)



X2: IO-digital & analog

DB15f HD	Function		HD15f HD
1	Analog In 1	Digital Out 3	9
2	Analog Out	Digital In 3	10
3	Out-Supply 24 V ext	Gnd	11
4	Digital Out2	Analog In2	12
5	Digital In2	Digital Out 4	13
6	5V	Digital In 4	14
7	Digital In return	Digital In 1	15
8	Digital Out 1		

X3: RS-232 In

1:1 connection

DB9f	Function		DB9f
1			6
2	TX		7
3	RX		8
4			9
5	GND		

X4: RS-232 Out

1:1 connection

DB9m	Function		DB9m
1			6
2	TX		7
3	RX		8
4			9
5	GND		

X5: Power

DC-Plug 2.1	Function
Center	12..24 VDC
	GND

Connector: DC-Plug coaxial, Center positive standard



Ensure that the power supply output voltage does not exceed the maximum input voltage of the controller (Umax=26 VDC)!

Wrong polarity of the power supply will damage the controller!

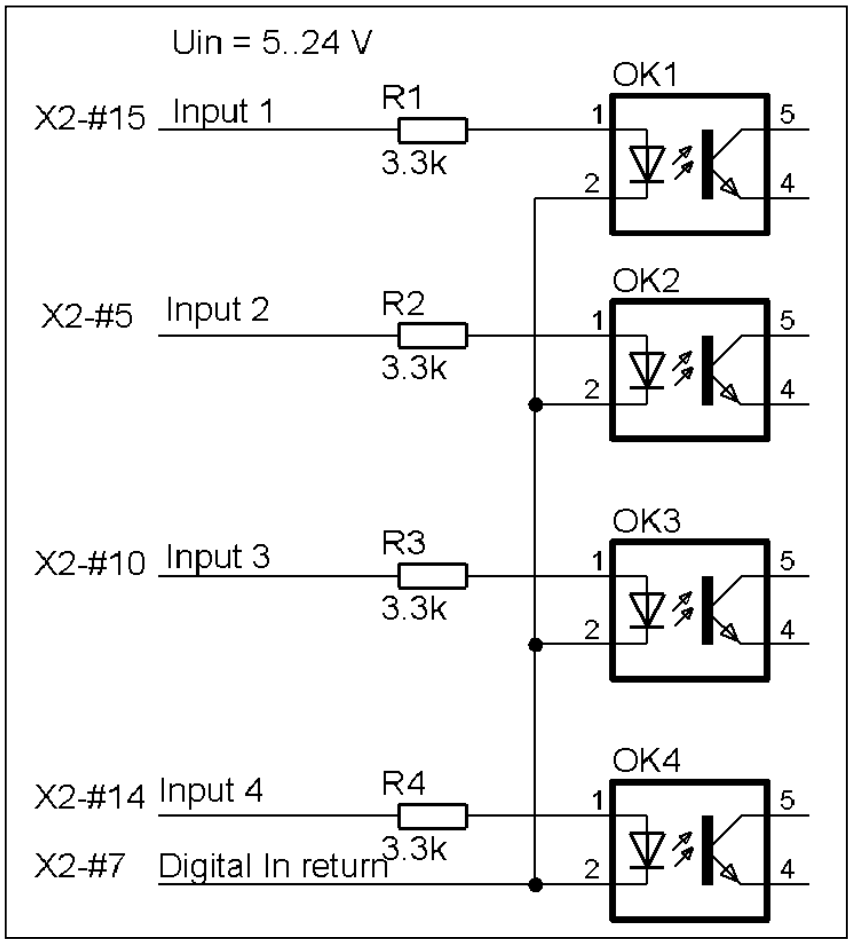
Status Display

In normal operation the status display shows the actual board address. If an error occurs, the error number is shown (see command TS – Error-Byte).

Digital / Analog IO :

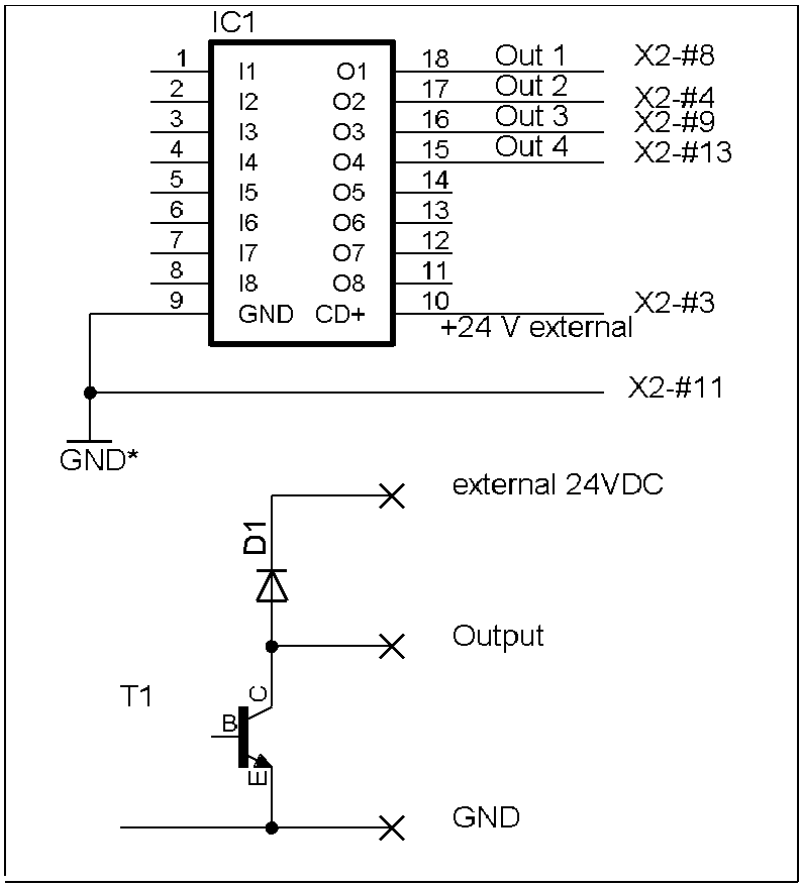
Digital-Inputs:

Number	4	
Type	optoisolated, sourcing	
Voltage	5..24 VDC	
Imax	<10 mA	



Digital-Outputs:

Number	4	
Type	Open-collector	
Voltage	24 VDC	
Imax	250 mA	per channel



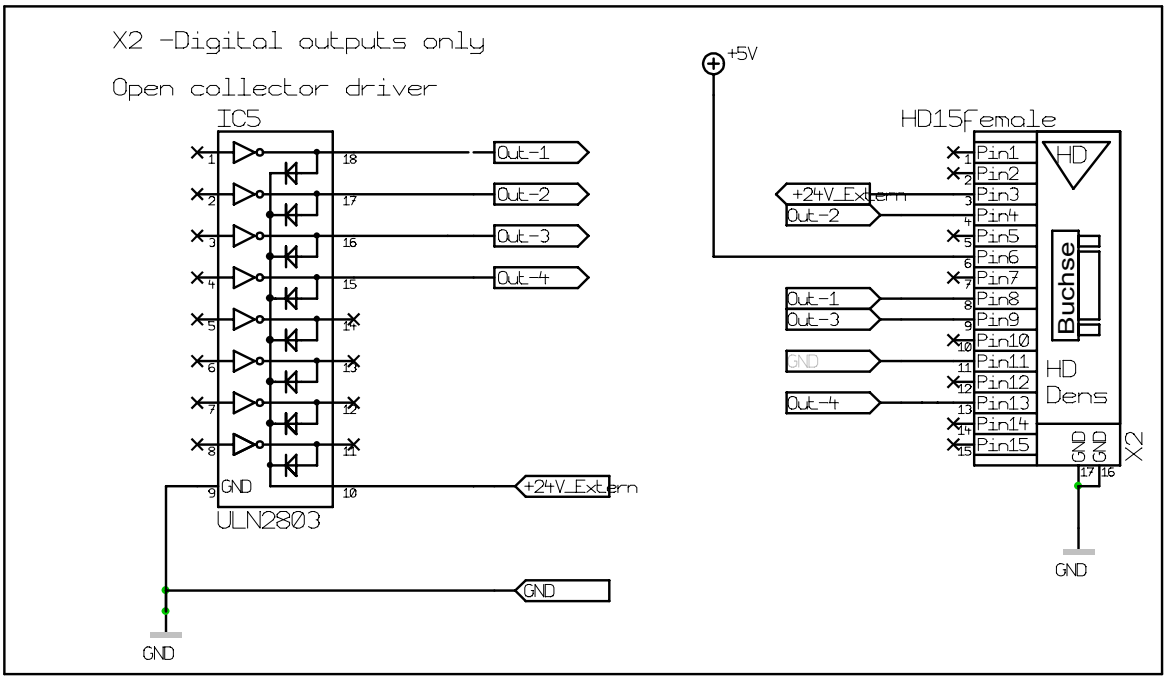
Using the outputs:

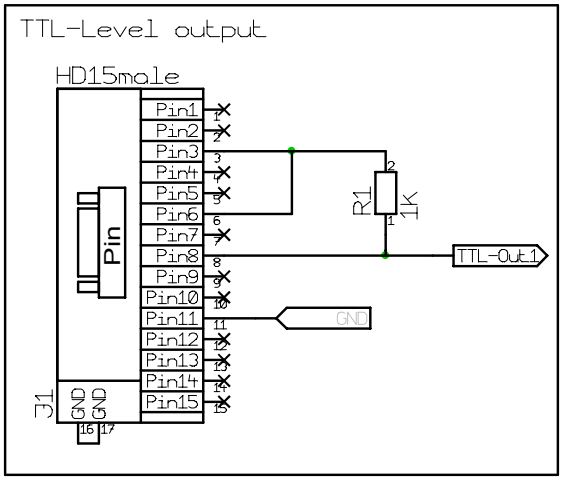
The ULN-2803 driver features s a continuous load current to 250 mA for each of the drivers, The high-current Darlington arrays are ideally suited for interfacing between low-level logic circuitry and multiple peripheral power loads. The open-collector outputs with integral clamp diodes are capable to drive resistive and inductive loads.

The load must be connected between *Output –pin* and *Out-Supply –pin*

Outputs to TTL-level:

Follwing shematic shows a simple adaption to TTL-level.





Analog-Output:

Number	1	
resolution	8 bit	
Voltage	0..5 V	
Imax	2 mA	

Analog-Input:

Number	2	
resolution	8 bit	
Voltage	0..5 V	

Jumpers:

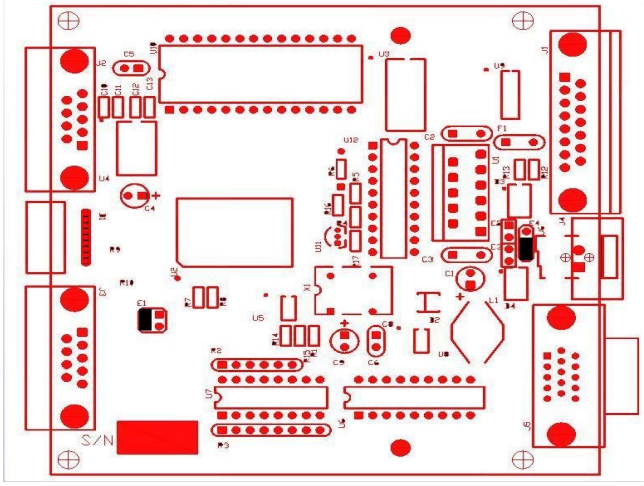
E1 don't change

E2 Place jumper here only to **disable the positive limit switch**. This jumper should be used only in special conditions such as a rotary table that has no mechanical limits of motion.

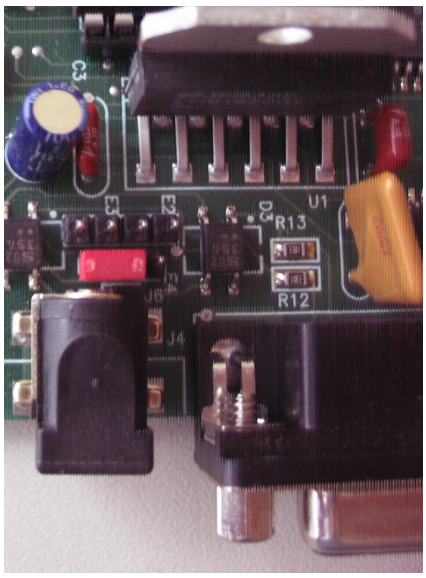
E3 Place jumper here only to **disable the negative limit switch**. This jumper should be used only in special conditions such as a rotary table that has no mechanical limits of motion.

E4 This jumper selects the return signal of the limit switch optoisolators. If the jumper is placed on the upper two pins, then the 24V motor supply is used (npn-limit function).

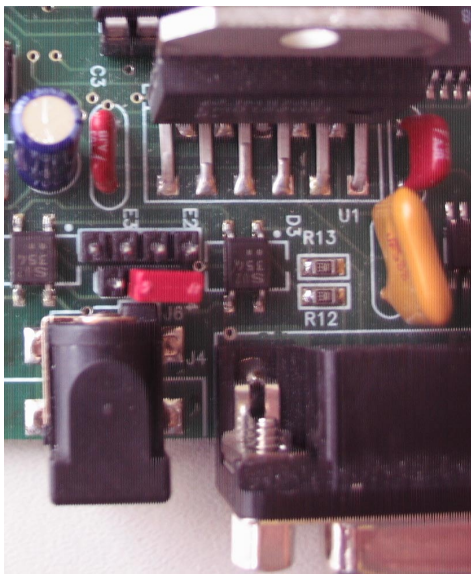
If the lower pins are jumpered, the system ground is supplied (pnp-limit function).



pnp-configuration

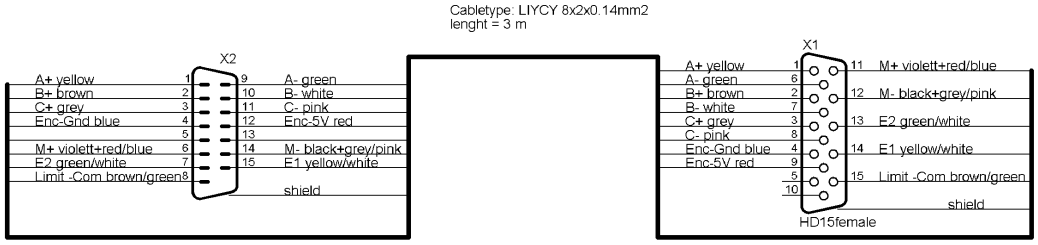


npn-configuration

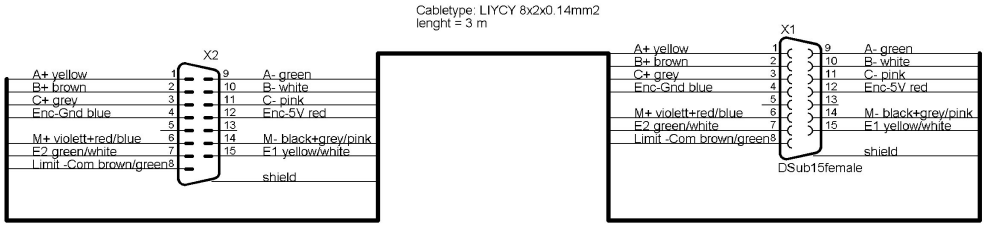


Motor-Cable:

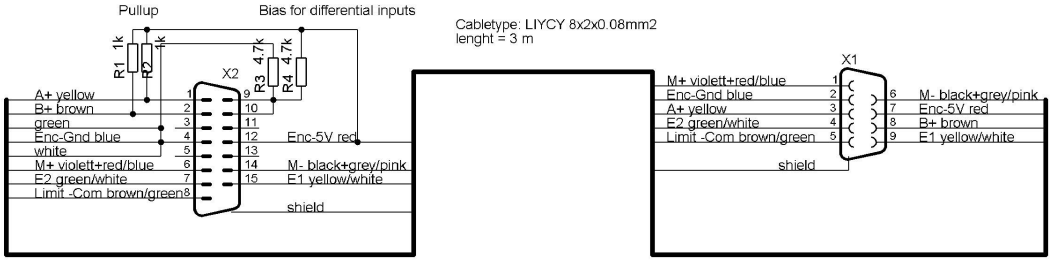
This is a schematic for a standard cable DB15male → DB15female HD **(PNP)**



This is a schematic for a standard cable DB15male → DB15female **(PNP)**



This is a schematic for a standard cable DB15male → DB9female. **(PNP)**
Encoder single ended, with bias resistors for the open inverted inputs (A- B-) and recommended pull-up resistors .



Please always use shielded and twisted-pair cables!

Communications

RS-232 Settings

Baudrate	**9600** / (19200 / 38400)
Bits	8
Stopbit	1
Parity	no
Protocol	no

How to communicate

The MoCo *series* has an internal command interpreter. All data to or from the controller are ASCII-strings. A command-string starts with a 2 letter mnemonic and dependent on the command, one or more parameters!
Over 50 commands are available for programming the MoCo.

Commands may be executed in various ways:

Single command -- One function executed immediately
Compound command -- Multiple functions executed immediately
Macro command -- Compound command stored for later execution
Sub-commands-- Single-character commands
Address-command—Network address selection

Command Termination

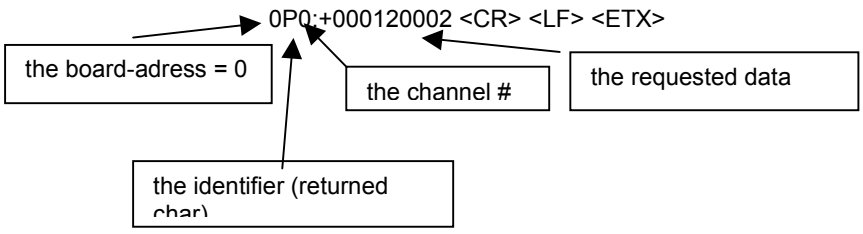
For communications to occur, first an existing board-address must be selected. (*see Chapter Address-Selection)* .
A command must be terminated by the *Carriage-Return Char = dec 13 (*<CR>)

A <CR> without any other character, repeats the last command!

Reply-Termination

All data sent from the MoCo *series* are ASCII Strings. An answer string starts with a command identifier which consists off:
the board-address,
a one character identifier (return char),
the channel-number,
a colon,
the requested data,
the reply termination CR LF ETX (dec 13, 10, 3 or hex D, E,3)

Example: TP<CR>



Single Command

A ***single command*** is executed immediately after the carriage return is received and will be repeated each time the carriage return is received, until a different command is entered. With this feature, it is very easy to continuously monitor the state of an input by simply holding down the "Enter" key.
Both uppercase and lowercase characters are valid, and spaces are allowed.

Compound Commands

A compound command is a series of single commands separated by commas rather than by a carriage return. In this way, it is possible to string together several commands before terminating with the carriage return. These multiple commands will be executed sequentially.

The syntax for a compound command is:
CMD[n], CMD[n], ...,..., (rtn)

Once this command is entered, it remains in the buffer until replaced by another command and can be re-executed by transferring a carriage return. Compound commands may contain up to 15 single commands.

Example:

ma0,ws,mr500<CR>
ma0,ws,mr2000,ws,rp10<CR>

Macro Commands

Macros can be a most powerful tool for the programmer. A macro command is a grouping of commands to form a short program, implemented by a macro number.
To use macros for programming the **MoCo** controller, insert an **MD** (Macro Definition) command as the *first* instruction in the command string. The syntax for macro commands is:

MD(macro#), followed by a compound command string.

Example: MD3,MR1000,WS500,MR-1000,WS,RP5 (rtn)

In this example, MD3 defines macro #3. To call up this macro, just issue the command MC3.
Macro commands may be stored in any order, but you may prefer to number them sequentially as they are entered, because the system gives no warning if you define (and overwrite) an existing macro. You may wish to do this under many conditions, such as when one macro is called by another. It is sometimes desirable to define a complex motion in one macro and define key parameters such as torque, gain, or velocity, in another macro which is called by the main macro.
Macro commands can call another macro, without limit. For instance, MC1 could call MC2, and MC2 could then call MC3 and still be able to return to complete the remainder of MC1.

Example: MD1,MC2,MC3,MC4,MC5,MC6

Macro commands may contain up to 15 single commands.

Sub Commands

Sub-commands may be used at any time. They are most useful for interrogating variables without disturbing an operating program. An example would be a situation where a repetitive motion is in process, such as dispensing adhesive in a pattern. The operator would like to know the status of the command without stopping it. The sub-commands can be used to read the number of iterations in a loop, current system status, position, etc. A single character is also provided for emergency stop action. If a macro is running standard commands are not possible witout interrupting the macros!

Sub-commands are one character commands without any command termination.

character	hex-value	decimal-value	standard-command	function
blank	h20	32		Stops a running macro
‘	h27	39	tp	returns the actual position
?	h3f	63	tf	returns the actual position-error
\	h5c	92	ms	returns Byte 1 of the commands TS
%	h25	35	ts	returns the status of the system 6 bytes
!	h21	33	ab	stops motion (all controllers)
#	h23	35	tc	returns the status of digital inputs/outputs

Control Characters:

character	hex-value	decimal-value	function
CR	$E	13	command /reply termination
LF	$D	10	reply termination
ETX	$3	3	reply termination
CtrlA	$1	1	address selection

Address Selection

Up to 16 controllers can share one serial interface. The address selection is done by the command *Ctrl+A* + the controller address, defined by command *<db>,* all others are automatically de-addressed.

Ctrl+A = ASCII table character 1

Note: After power-up, the controller is not addressed! Send first the address command!

Example:	CtrlA5	address Controller #5
	tp	query position
	5p0:+000300	reply of controller #5
	CtrlA2	address Controller #2
	tp	query position
	2p0:+0002300	reply controller #2

There is a interesting feature which enables sending commands to all available controllers.
The command *Ctrl-a+v+ standard command*
Please do not use this command with reporting commands, the data will be chaotic!

Example:	CtrlAV DV2000	defines the velocity of all boards
	CtrlAV MR2000	all motors will move 2000 inc relative

Limit Switch Operation

Limit switch enable/disable

The limit switch operation in MoCo controllers is somewhat complex, in order to give the maximum protection in all circumstances. The complexity arises due the the extreme variation in conditions that must be protected.

First and foremost is the hardware protection. Whenever the limit switch is activated that is in the direction of travel, the drive to the motor is disabled within a few nanoseconds after receipt of the signal from the switch. This gives absolute protection against damage by overtravel and is not affected in any way by software operation.

However, there are applications involving the use of rotary stages where there is no need for limit switch protection and where disabling the motor drive might cause a problem. For these applications, beginning with the MoCo2006 boards, (S/N 0610-xxx and 0611-xxx) there is now the ability to disable either or both limit switches using jumpers at E2 and E3. (No jumper = limit switch enabled)

At the same time, the switches are still monitored by the software protection scheme, which is designed to reset the motion target to the actual position whenever either limit switch is activated. This action prevents continued motion if the limit switch were to be only a momentary activation. The software protection scheme only tests the state of the limit switches once per millisecond.

Whenever either or both limit switches are disabled by placement of the disable jumpers at E2 and E3, the corresponding software limit should be disabled. In previous releases, the LN (limit on) and LF (limit off) commands addressed both limit switches. Both switches are still controlled when the LN and LF commands are given with no following argument, to retain compatibility with previous firmware revisions. However, it is now also possible to enable or disable the limit switches separately. LN1/LF1 enables/disables only the negative limit switch and LN2/LF2 enables/disables only the positive limit switch.

Even with the limit switches disabled, it is still possible to make use of the negative limit switch as a reference input. The FE1 (Find edge command, mode 1) command causes motion in the negative direction until the negative limit switch is activated. Normal move commands will not be stopped at this position, so it is important that only the FE1 command be used to locate the switch position if the intent is to establish a reference position. There is no corresponding use available for the positive limit switch.

After stopping, the FE4 (Find edge command, mode 4) command may be used to move in the positive direction until the limit switch is deactivated. It is possible to use this process to establish a very repeatable reference position.

When using the negative limit switch input only as a reference switch, the corresponding limit switch disable jumper should be installed, as well as the LF1 command being given.

Switch polarity

The LH and LL commands are provided so that limit switches may be supported, regardless of the active state. If the switch input is a logic high level when the switch is activated, then the LH (Limit high) command should be used. Otherwise, use LL (Limit low). After selecting the active state, the UD (Update) command should be used to save the setup so that it does not have to be repeated for later use.

Please be aware that the UD command saves all parameter setups, so if a gain has been changed for testing, the changed value will now become the value at all subsequent uses, until the gain value is changed and the UD command saves the changed value.

The firmware for Rev. 3.0 and later always report the limit switch state as a “1” when activated, regardless of the LH/LL state.

Command Reference

Data formats:

l	integer (24 bit)
ui	unsigned integer (24 bit)
w	word (16bit)
b	byte (8 bit)

tp Tell Position

Description	Tell Position reports the absolute position in increments of the encoder. **TP** may be used to monitor motion during both motor on and motor off status.
Parameter	no
Return Char	P
Index	always 0
Return Value	l [increments]
Example	tp 0P0:+0000020002

tt Tell Target

Description	Reports target position in encoder increments. This is the absolute position to which the servo loop will try to drive the motor any time the **MN** (Motor ON) state is in effect. The target position may be specified directly with the **MA** (Move Absolute) and several other commands, or indirectly with the **MR** (Move Relative) command. If the system is in decimal mode, ten digits will be reported with a leading minus sign (-), if the position is less than the position defined as "home."
Parameter	no
Return Char	T
Index	always 0
Return Value	l [increments]
Example	tt 0T0:+0000020002

ts

Tell Status

Description

When the **TS** command is given, the status of the system as well as the motion and limit switches is reported. The answer is a string with 7 HexBytes (B_0 .. B_6)
If an error occurs (LED Display) the error will be reset by this command!

Parameter no
Return Char S
Index always 0

b0 b1..b2 b3 b4 b5 b6

Byte_0: LM629 Status-Byte

Bit 0	Busy
Bit 1	Command error
Bit 2	Trajectory complete
Bit 3	Index-found
Bit 4	Position-limit exceeded
Bit 5	Excessive position error
Bit 6	Breakpoint-reached
Bit 7	Motor off

Byte_1: Operational Flags 1

Bit 0	
Bit 1	Wait in process
Bit 2	Command error
Bit 3	Leading zero suppression active
Bit 4	Macro command called
Bit 5	Leading zero suppression disabled
Bit 6	
Bit 7	Echo on

Byte_2: Operational Flags 2 (only for factory use)

Bit 0	
Bit 1	
Bit 2	
Bit 3	
Bit 4	
Bit 5	
Bit 6	
Bit 7	

Byte_3: Motion Flags

Bit 0	Breakpoint in process-flag
Bit 1	Polarity of dynamic error
Bit 2	Polarity of move direction 1= positive
Bit 3	Move Error
Bit 4	Polarity of move at start of deceleration
Bit 5	Velocity mode
Bit 6	Excessive position error
Bit 7	LM629 access in process

Byte_4: Limit Switch Status

Bit 0	Limit forward Switch enable flag (1=active)
Bit 1	Limit Switch active state (1 = high)
Bit 2	Find Edge in process
Bit 3	Brake on Flag (bn = 1 bf = 0)
Bit 4	Find limit switch flag
Bit 5	Ref-switch polarity control
Bit 6	Limit reverse switch enable flag (1=active)
Bit 7	

Byte_5: Limit Switch inputs

Bit 0	Limit reverse active (1=active)
Bit 1	Limit Forward disabled by jumper E2 (disabled =0)
Bit 2	Limit forward active (1=active)
Bit 3	Limit reverse disabled by jumper E3 (disabled =0)
Bit 4	
Bit 5	
Bit 6	
Bit 7	

Byte_6: Error Codes

01	command not available
02	first command character must be a letter
03	no command
04	negative not allowed
05	character following command must be a number
06	value too large
07	value too small
08	continuation character must be a comma
09	command buffer overflow
0A	macro storage overflow

Hint:

For proper sequencing of motion Byte 0 Bit 2 is the most interesting bit. Bit 2 is the **trajectory complete interrupt flag**, this is set to logic one when the trajectory programmed and initiated by any motion command has been completed. Because of overshoot or a limiting condition (such as commanding the velocity to be higher than the motor can achieve), the motor may not yet be at the final commanded position. It strongly depends on dynamic and control settings, it's always a good idea to add some time after a move until the system is settled. Polling the position error (**tf**) could also help.

Typical values for settling time could be 50-300 ms.

ms Move Status

Description: Reports only Byte 0 of the **TS** command

Parameter: no
Return Char: M
Return Value: b
Index: always 1
Example: ms
0M1:4C

tf — Tell Following Error

Description	Reports the difference between the dynamic target and the actual position. During motion, it is normal for the actual position to lag behind the target position by some amount, usually dependent on the programmed velocity. If the velocity is higher than physically possible for the system, or if it has encountered an impediment, the following error will increase. If the obstruction is temporary, the servo action will attempt to restore the error to zero when it is removed.
Parameter	no
Return Char	F
Index	always 0
Return Value	I [increments]
Example	tf 0F0:+0000000022

tb — Tell Board Address

Description	Reports the board-adress
Parameter	no
Return Char	B
Index	always 0
Return Value	number 0.. F
Example	tb 0B0:01

db — Define Board Address

Description	Defines the board-adress
Parameter	the address
Return Char	no
Index	no
Return Value	no
Example	db5

ud

Update

Description	Stores all parameters like velocity and filter parameters in the non-volatile memory. When the MoCo powers up these parameters will be active!
Parameter	no
Return Value	no
Example	ud

dv

Define Velocity

Description	Defines the desired velocity for any movement. Causes the motor to run at velocity ***n*** in subsequent motion command***s.*** per second. If the torque load changes on the motor, the controller attempts to maintain the velocity by varying the motor current. The value ***n*** may be in the range from 1000 to 255,000.q However, the usable range of velocity settings is determined by the number of lines in the encoder and the maximum RPM of the motor in use. Many systems will fall into a range of 50,000 to 250,000 counts/second.
Parameter	l [increments/s]
Parameter-Range	0..1 000 000
Return Value	no
Example	dv20000

gv

Get Velocity

Description	Returns the programmed velocity, (defined with command *dv*)
Return Char	Y
Index	always 0
Return Value	l [increments/s]
Example	gv 0Y0:+0000020000

da

Define Acceleration

Description	Defines the desired acceleration **i** for any movement. The maximum value is mostly limited through the mechanical system (the inertia of the system)
Parameter	I [icrements/s^2]
Parameter-Range	0..10 000 000
Return Value	no
Example	da200000

ga

Get Acceleration

Description	Returns the programmed acceleration, (defined with command *da*)
Return Char	A
Index	always 0
Return Value	I [icrements/s^2]
Example	ga 0L0:+0000200000

dp

Define P-Term

Description	Defines the P-Term of the PID-filter. This command sets the slope of the proportional relationship between the position error and the motor voltage. The higher the gain value is set, the higher is the stiffness of position coupling, so that an small error value causes a proportionally larger motor current driving the motor towards the target. The default gain value is usually stable. The optimum value depends on friction, inertia, motor power, and the resolution of the encoder. It must be determined by the user. If the error reported by an axis after completing its motion is excessive, the gain value may be increased in small increments until the error is within acceptable limits. If the axis becomes unstable and begins to oscillate, the gain must be reduced until the oscillation stops. If the error remains too high, use the *di* command.
Parameter	w
Parameter-Range	0..32767
Return Value	no
Example	dp800

gp

Get P-Term

Description	Returns the programmed p-term, (defined with command *dp*)
Return Char	G
Index	always 0
Return Value	w
Example	gp 0G0:+0000000800

di

Define I-Term

Description	Defines the I-Term of the PID-filter. Sets the gain to be applied to the integral term in the PID algorithm. The primary function of this term is to overcome friction-induced static errors. Note: The I-term has no effect unless the dl command is used to set the limit of output>0
Parameter	w
Parameter-Range	0..32767
Return Value	no
Example	di200

gi

Get I-Term

Description	Returns the programmed integral-term, (defined with command *di*)
Return Char	I
Index	0
Return Value	w
Example	gi 0I0:+0000000400

dd

Define D-Term

Description	Defines the D-Term of the PID-filter. Sets the gain to be applied to the derivative term in the PID algorithm.
Parameter	w
Parameter-Range	0..32767
Return Value	no
Example	dd200

gd

Get D-Term

Description	Returns the programmed derivative-term, (defined with command *dd*)
Return Char	D
Index	0
Return Value	w
Example	gd 0D0:+0000000400

dl

Define Integral-limit

Description	Defines the integral-limit of the PID-filter.
Parameter	n
Parameter-Range	0..32767
Return Value	no
Example	dl2000

gl

Get Integral-Limit

Description	Returns the actual programmed integral-limit, (defined with command *dl*)
Return Char	L
Return Value	w
Index	always 0
Example	gl 0L0:+0000002000

de

Define Following-Error

Description	Sets the **maximum allowable** error between the dynamic target and the actual position. May be changed as often as desired to provide maximum protection to the system. The following error can be monitored with the TF command. For maximum system safety, use the *DE* command to limit following error to a value slightly above that required for normal operation. Whenever the following error exceeds the set limit, an internal MF command is given, which will prevent the use of any motion commands until the MoCo is reset or until an MN command is given.
Parameter	w
Parameter-Range	0..32767
Return Value	no
Example	de2000

br

Define Baud-Rate

Description	The MoCo is able to communicate with 3 standard Baud-Rates 0 : 9600 1: 19200 2: 38400 Caution: Parameter is not storable, after reset 9600 is always active!!
Parameter	b
Parameter-Range	0..2
Return Value	no
Example	br0 ; 9600 is active

rt

Reset Controller

Description	Restarts the internal firmware operation as if from a power off condition. All default values are restored. If Macro 0 exists, it will be executed.
Parameter	no
Return Value	no
Example	rt

ef

Echo Off

Description	Disables echoing. When control is from a computer program, it is sometimes easier to program if there is no echo, unless the program uses it for verification of successful transmission.
Parameter	no
Return Value	no
Example	ef

en

Echo On

Description	Enables echoing of command characters as they are entered. Each character received is echoed unchanged. This is a very useful feature when the MoCo is being controlled manually from a terminal..
Parameter	no
Return Value	no
Example	en

mn

Motor On

Description	This is the normal system control mode, where the MoCo controls the axis position continuously. Any deviation between actual and target position causes the motor to be driven toward the target, using the previously defined values of velocity, acceleration and PID filter parameters. When the MN command is given, the equivalent of an AB command is also executed, to prevent unwanted motion that could result from motion during an MF condition or from other sources. For example, whenever the position loop error exceeds the maximum setting as defined by the DE command, the motor loop is opened and it is necessary to give the MN command to regain normal control..All motion commands given after an excess error fault occurs will be ignored until the MoCo has been reset or until an MN command is given
Parameter	no
Return Value	no
Example	mn

mf

Motor Off

Description	When this command is issued, the motor is no longer held in position control and may be moved freely. The *MF* command is used to prevent unwanted movement or to allow for manual positioning of the unit. However, the motor position is always monitored, even in the ***MF*** status, and may be reported by the ***TP*** command.
Parameter	no
Return Value	no
Example	mf

ab

Abort Motion

Description	This command stops a motion ! The target position is changed to be equal to the present position. n=0 stop abrupt n=1 deccelerated stop
Parameter	n
Parameter-Range	0 ..1
Return Value	no
Example	ab1 decelerated stop

mr

Move Relative

Description	This command generates a motion of relative distance of ***n*** counts in the specified direction from the actual motor position. **i** may be either a positive or negative number up to a total target position +/-1 073 741 843
Parameter	I
Parameter-Range	depends on the absolute target, should not exceed the maximum position +/-1 073 741 843 !
Return Value	no
Example	mr5000

ma

Move Absolute

Description	This command generates a motion to the absolute position ***i.*** The zero, or home position, may be defined by the **DH** (Define Home) statement. If not otherwise defined, it is the position where the controller was when powered on.
Parameter	I
Parameter-Range	+/-1 073 741 843
Return Value	no
Example	ma2000

rp

Repeat

Description	This command causes the command string to repeat ***I+1*** times. If ***i*** is not specified, the commands are repeated 65,535 times. The repeat loop may be interrupted by transferring any character. This character may not be the first character of a new command because it will be discarded. (To repeat forever, use two **RP** commands in sequence.)
Parameter	n
Parameter-Range	0..65535
Return Value	no
Example	ma0,ws,ma2000,ws,rp99 (performs 100 iterations)

fe

Find Edge

Description	This command is used to initialize the system at a given position. The motor runs at a programmed speed until a change of state occurs on the limit-input line. The directionExample: **F**E0 (rtn) : Causes motor to move in a positive direction until the E2 input is activated. **F**E1 (rtn) : Causes motor to move in a negative direction until the E1 input is activated. **F**E4 (rtn) : Causes motor to move in a positive direction until the E1 input is deactivated.
Parameter	n
Parameter-Range	0,1,4
Return Value	no
Example	fe1 seach for E1 in negative direction

fi

Find Index

Description	This command is used to define a very exact homing-position. The motor runs at the programmed speed until the index of the encoder occurs. Bit 3 indicates if the index occurs (status & $08) Normal rotary encoders have one index per motor revolution!
Parameter	
Parameter-Range	
Return Value	no
Example	fi

dh

Define Home

Description	Defines the current motor position as zero position (home position).
Parameter	no
Return Value	no
Example	dh

gh

Go Home

Description	The Go Home command causes the motor to move to absolute zero position. Equivalent to an **MA0** (Move Absolute axis a to zero) command.
Parameter	no
Return Value	no
Example	gh

ws

Wait Stop

Description	Wait until axis has reached the end of the actual movement, and waits another *n* milliseconds before the execution of the next command. Without parameter the wait-time is 1000 ms!
Parameter	n
Parameter-Range	0..65535
Return Value	no
Example	mr2000,ws500,ma0

md

Macro Definition

Description	This command is used to define a new macro command. Any duplication of numbers will simply result in the loss of any previously defined macro using that number. To define a macro, choose any number in the allowable range for the new macro and enter MD followed by this number and a comma before entering the function you wish the macro to perform, in the normal manner. Macro 0 is an autostart macro, executed after booting the controller
Parameter	macro-number + command-string
Parameter-Range	0..14 (macro-number)
Return Value	no
Example	md1,ma0,ws,ma2000,ws

mc

Macro Call

Description	This command may be used to implement a previously defined macro command. If there is no macro defined by the number n, no action will be taken. **A running macro can be stopped by sending the <space>** dec=32 hex =$20
Parameter	n, macro-number
Parameter-Range	0.14
Return Value	no
Example	mc1

tm

Tell Macro

Description	Displays all previously stored macro commands. If ***n*** = 0 or, if ***n*** is not specified, all macros will be displayed. Since macros may be defined in any sequence, the **TM** command is useful for confirming the existence of, as well as displays all previously stored macro commands. If ***n*** = 0 or, if ***n*** is not specified, all macros will be displayed. Since macros may be defined in any sequence, the **TM** command is useful for confirming the existence of, as well as
Parameter	n, macro-number 0..14
Return Value	the contents of the macro(s)
Example	tm1 mc1,mr555,ws,gh

rm

Reset Macro

Description	Resets macro-memory
Parameter	0..14macro-number)
Example	rm reset all macros rm 1 reset only macro 1

so — Set Analog Output

Description	Sets the analog output to a value. The output is a 0 to 5 V 8 bit output. One bit is 19.5 mV
Parameter	b -> value of the output 8 bit 0..255
Example	so255 the output is 5 V

io — Increment Analog Output

Description	Increments the analog output . The output is a 0 to 5 V 8 bit output. One bit is 19.5 mV.
Parameter	I -> incremental value -255..255
Example	io2 ; new analog value = old value + 2 mc2,io1,ws1,rp ; generates a analog sawtooth

to — Tell Analog Output

Description	Returns the actual value of the analog output
Parameter	no
Return Char	o
Index	always 0
Return Value	b [0..255]
Example	to 0o0:255

ta

Tell Analog Input

Description	Returns the current value of the analog inputs 1 and 2
Parameter	b channel [0 ..2] 0 = report both inputs 1..2 = reports channel #
Return Char	A
Index	b [0..2]
Return Value	n m -> value of the analog-inputs 8 bit n = channel 1 m = channel 2 0..255
Example	ta 0A0: 127 230 ta1 0A1 :127 ta2 0A2 :230

tc

Tell Channel

Description	Returns the current status of the digital-inputs and outputs.
Parameter	b [0..4] 0 = report all outputs and inputs 1..4 = report input-channel #
Return Char	H
Index	0 ..4
Return Value	n m -> value of the inputs /outputs 8 bit n = outputs m = inputs
Example	tc 0H0: 0 255 ; all outputs and inputs not active tc1 0H1: 1 ; reports channel 1

cf — Channel oFf

Description	Reset digital output(s)
Parameter	b, 0..4 0 = all outputs 1..4 = only channel #
Return Value	no
Example	cf 0 or cf resets all 4 digital outputs cf1 resets channel 1

cn — Channel oN

Description	Set digital output (s)
Parameter	b, 0..4 0 = all outputs 1..4 =only channel #
Return Value	no
Example	cn0 or cn sets all 4 digital outputs cn1 sets channel 1

wn — Wait oN

Description	Stops Command Execution until Input Channel on
Parameter	b [1..4] 1..4 = input-channel #
Return Char Index	1 ..4
Return Value	no
Example	wn1,ma1000

wf — Wait oFf

Description	Stops Command Execution until Input Channel off
Parameter	b [1..4] 1..4 = input-channel #
Return Char	
Index	1 ..4
Return Value	no
Example	wn1,ma1000,wf1,ma0

xn — eXecute if oN

Description	Executes Commands if Input-Channel on
Parameter	b [1..4] 1..4 = input-channel #
Return Char	
Index	1 ..4
Return Value	no
Example	xn1,mr555,ws10 move if Input1 = on xn1,xn2,mr599,ws10 move if Input 1 & 2 = on

xf — eXecute if oFf

Description	Executes Commands if Input-Channel off
Parameter	b [1..4] 1..4 = input-channel #
Return Char	
Index	1 ..4
Return Value	no
Example	xn1,mr555,ws10 move if Input1 = on xn1,xn2,mr599,ws10 move if Input 1 & 2 = on xn1,xn2,xf3,mr599,ws10 move if In 1 & 2 = on & In 3 = off

Trigger Functions:

The Moco controller is able to generate position synchrone trigger outputs. This helps to synchonize lasers or other measurement devices.
Following limitations have to be considered if using this option:

- Multible triggers are limited to a maximum frequency of **500 Hz**!
- The accuracy of the trigger output is directly dependent on the actual velocity . There is a time delay of about 1 ms (reaction time of a interrupt including processing time to link to the output)

→ the error is app. 1/1000 of the actual velocity in counts/sec

Example: The application needs every 5 µm a trigger signal. The first trigger should occur at absolute position 10 µm.
1µm = 2 encoder-increments
The output for the trigger is Out 1, the pulswidth = 2.5 µsec

```
actual position is : 0
sb1
pw0
bd20
bi10
mr500
```

Maximum speed should be: 500 [1/s] x 10 [increments] = 5000 [Increments/s]

sb SetBreakpoint port

Description	Assignes the trigger-output port (digital output 1..4)
Parameter	b, 1..4 1..4 = port number
Return Value	no
Example	sb1 ; defines output 1 as trigger output

bd Breakpoint Define

Description	Defines the absolute position for the first breakpoint Please note, the stage should be a little before the first trigger! It also sets bit 0 of the 4th byte reported in TS
Parameter	I +/-1 073 741 843
Return Value	no
Example	ma19900,sb1,bi100,bd20000,ma30000 first trigger at position 20000, next every 100 increments

bi

define Breakpoint Increment

Description	Defines the relative distance (breakpoint increment) between the trigger-points.
Parameter	I +/-1 073 741 843
Return Value	no
Example	ma19900,ws,sb1,bi100,bd20000,ma30000 first trigger at position 20000, next every 100 increments ma20100,ws,sb1,bi-100,bd20000,ma0 first trigger at position 20000, next every -100 increments

pw

define Pulse Width

Description	Defines the puls-width of the trigger signal 0= 2.5µsec 1..255 = 1..255 ms
Parameter	b 0..255
Return Value	no
Example	pw1 ; puls width = 1ms

Some examples:

Homing

DV10000	Slow velocity to search the E1 switch
FE1	search reverse direction
WS	Wait until after move complete 1000 ms
DV500	very slow velocity
FE4	search for toggle of the E1 switch in forward direction
WS	Wait until after move-complete 1000 ms
DH	Define-home, actual position = 0

Demo

MA0	Move absolute to 0
WS500	Wait until after move-complete 500 ms
MR500	Move relative 500 increments
WS	Wait until after move-complete 1000 ms
RP49	repeat the line 50 times

Some more macros

Macro 2: This macro calls other macros

MD2, EM+0000000010,EM+0000000011,EM+0000000012,MA,WS+0000000500,RP

Macro 10: motor moves 20000 increments relative, after trajectory complete, output 1 goes high for 200 ms, the output goes low

MD10,MR+0000020000,WS+0000000001,CN+0000000001,WA+0000000200,CF+0000000001,WS+0000000001

Macro 11: motor moves 500 increments relative, after trajectory complete wait for 100 ms , repeats 11 times this macro

MD11, MR+0000000500,WS+0000000100,RP+0000000010

Macro 12: same like macro 11 in negative direction

MD12, MR-0000000500,WS+0000000100,RP+0000000005

Trouble Shooting:

No contact with the controller:

- interface settings correct?
- RS232 connection correct ?
- Controller addressed?
- 2 or more controllers with same address?

Motor without power or do not move:

- motor connected ?
- motor-off due excessive positioning error ?
- after booting the controller motor always off
- limit switch active in direction of movement?

Excessive Positioning Error:

- motor not or wrong connected
- encoder not or wrong connected
- speed too high
- acceleration too high
- torque / force too high

Command Short Form

read-command	reply-char	set-command
TS **T**ell **S**tatus	S:	
MS **M**ove **S**tatus	M:	
HE Print list of commands		
TE **T**ell **E**rror (distance from target)	E:	
TP **T**ell **P**osition	P:	
TT **T**ell **T**arget	T:	
TF **T**ell **F**ollowing error (distance between position and dynamic target)	F:	
TD **T**ell **D**ynamic target	N:	
TV **T**ell actual **V**elocity	V:	
GV **G**et**V**elocity	Y:	DV **D**efine **V**elocity
GA **G**et**A**cceleration	A:	DA **D**efine **A**cceleration
GP **G**et **P**roportional gain	G:	DP **D**efine **P**roportional gain
GI **G**et **I**ntegral gain	I:	DI **D**efine **I**ntegral gain
GD **G**et **D**erivative gain	D:	DD **D**efine **D**erivative gain
GL **G**et Integral-Limit	L:	DL **D**efine integral **L**imit
		DE **D**efine maximum following **E**rror
VE Display **VE**rsion number		
TI **T**ell the **I**teration number	X:	
TB **T**ell **B**oard address	B:	DB Define board-adress
CS Perform self-test checksum	C:	
		LN **L**imit switch operation o**n**
		LF **L**imit switch operation o**ff**
		LL **L**imits switches active **l**ow
		LH **L**imit switches active **h**igh
		UD **U**p**d**ate
		DM Input and output in **D**ecimal for**m**at
		HM Input and output in **H**ex for**m**at
		RT **R**eset all parame**t**ers to default and do power-up start
		EF Turn o**ff** **e**cho to RS-232 port
		EN Turn o**n** **e**cho to RS-232 port
		AB **Ab**ort Motion
		MN **M**otor O**n**
		MF **M**otor o**ff**
		WA **Wa**it specified time
		MR **M**ove **R**elative
		MA **M**ove **A**bsolute
		DH **D**efine **H**ome
		GH **G**o **H**ome
		RP **Rep**eat from beginning of line
		WS **W**ait until end of trajectory
		FE **F**ind **E**dge
		FI Find Index
TM **T**ell **M**acro (1..31) TZ **T**ell Macro **Z**ero		MD **M**acro **D**efinition
		MC **M**acro **C**ommand (or EM)
		RM **R**eset **M**acro
		RZ **R**eset **M**acro 0
TO **T**ell Analog **O**utput	O	SO **S**et Analog**O**utput
TA **T**ell **A**nalog Input	A	
TC **T**ell **C**hannel	H	CF **C**annel **O**ff
		CN **C**annel o**N**

Factory Settings :

DV **D**efine **Velocity**	
DA **D**efine **A**cceleration	
DP **D**efine **P**roportional gain	
DI **D**efine **I**ntegral gain	
DD **D**efine **D**erivative gain	
DL **D**efine integral **L**imit	
DE **D**efine maximum following **E**rror	
EN Turn o**n** **e**cho to RS-232 port	